

Interview Questions — Capstone Project

Total Questions: 43

1. What is the main purpose of a capstone project in a React JS course?

Hint: Think about how a capstone project brings together multiple skills.

2. How would you explain your project's scope and main goals to a non-technical client?
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3. What is the difference between brainstorming and finalizing a project idea?

Hint: One is about generating possibilities, the other is about making a decision.

4. Give an example of how you could use ChatGPT to help brainstorm features for a React-based project.
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5. What is the difference between low-fidelity and high-fidelity wireframes?

Hint: Consider the level of detail and purpose of each type.

6. How would you use Figma AI or Uizard to speed up your prototyping process for a dashboard UI?
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7. What does user journey mapping mean, and why is it important when designing a UI?

Hint: Think about how users interact with your app from start to finish.

8. Describe the steps you would take to create a wireframe for a homepage using an online tool.
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9. When setting up a new React or Next.js project, what folder structure would you choose to keep the project scalable?

Hint: Consider separation of concerns like components, pages, assets, and services.

10. Why is it important to use environment variables (like .env.local) for API keys in a React project?
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11. How would you securely store and access your OpenAI or Firebase keys in your React project?
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12. What steps would you follow to set up a basic React app and connect it to Firebase?
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13. Explain the difference between a React component's props and state.

Hint: Think about where the data comes from and who controls it.

14. Write a simple React functional component that displays a card with a title and description passed as props.

15. How would you make a navigation bar component reusable across multiple pages in your project?

16. What are the advantages of designing modular, reusable React components?

Hint: Consider code maintenance and scalability.

17. How could you use GitHub Copilot to help you scaffold repetitive UI components in React?

18. Explain how you would connect a React form to Firestore to store and fetch user data.

19. What are CRUD operations, and how do you implement them using Firebase or a GraphQL API in a React app?

20. Write a code snippet to fetch data from Firestore and display it in a React component.

21. If an API call in your React app is failing, how would you troubleshoot and improve the error handling?

Hint: Think about `async/await` and `try/catch` blocks.

22. Describe how authentication works using Firebase Auth or NextAuth.js in a React or Next.js project.

23. What is the role of Redux or Context API in session management for user authentication?

Hint: Think about how user state is shared across components.

24. How would you implement a login form and persist the user's session in your React project?

25. What is the difference between Actions, Reducers, and Store in Redux?

Hint: Focus on the flow of data and state updates.

26. Write a simple Redux reducer to handle user login and logout actions.

27. How can Redux DevTools help you debug state changes in your React application?

28. What are the benefits and challenges of managing complex state using Redux compared to Context API?

29. Describe how you would integrate the OpenAI API to generate content (like a blog summary or resume section) in your React app.

30. How does prompt tuning affect the quality of AI-generated content from APIs like OpenAI?

Hint: Consider how changing the input affects the output.

31. Write a function in React that sends user input to the OpenAI API and displays the AI-generated response.

32. How would you test your React app's design for responsiveness across mobile, tablet, and desktop devices?

Hint: Think about browser DevTools and media queries.

33. What is Google Lighthouse, and how can it help you optimize the performance of your React project?

34. If your Lighthouse report shows poor performance, what are some steps you could take to improve it?

35. Explain how you would check browser compatibility for your React application.

36. Describe the process of deploying your React or Next.js project to a platform like Firebase Hosting, Netlify, or Vercel.

37. What steps are involved in connecting a custom domain to your deployed project?

38. Briefly explain what CI/CD is and why it might be useful for future projects.

39. How would you combine multiple reducers in Redux, and why is this useful in larger projects?

40. Describe a scenario where you would use middleware in Redux, and what problem it solves.

41. How could you use ChatGPT or Copilot to help you debug a tricky state management issue in your React app, and what would you do to make sure the AI's suggestion is actually correct?

Hint: Think about testing, code review, or verifying the logic yourself.

42. If you used Figma AI or Midjourney to help design your UI, what would you still need to check or adjust manually before finalizing your project?

Hint: Consider usability, brand consistency, and technical limitations.

- 43.** Imagine you built an AI-powered blog creator using React, Firebase, and the OpenAI API. Walk me through how the frontend, backend, and AI integration work together in your project.
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